PRACTICAL 9

Aim

To perform social media trend detection using Apache Spark with PySpark by analyzing hashtag frequency, engagement, categories, regions, and daily trends.

The practical focuses on creating a social media hashtag dataset, loading the data into a Spark DataFrame, examining its schema and records, calculating hashtag frequency, identifying top trending hashtags, analyzing trends by category and region, comparing trends across multiple days, generating a trend analysis report, and visualizing hashtag trends.

Problem Definition

Social media platforms such as Twitter/X, Instagram, Facebook, and LinkedIn generate a massive amount of data every day. Users continuously post content related to current events, technology, sports, entertainment, politics, business, and other topics.

A social media analytics company needs to identify popular and emerging topics from this large volume of data. Traditional data processing tools may become inefficient when processing large-scale social media datasets.

Apache Spark provides distributed DataFrame operations and aggregation functions that can efficiently process large datasets. Spark can be used to analyze hashtag frequency, engagement, categories, regions, and trends over time.

The task is to create or obtain a social media hashtag dataset and perform trend detection using PySpark.

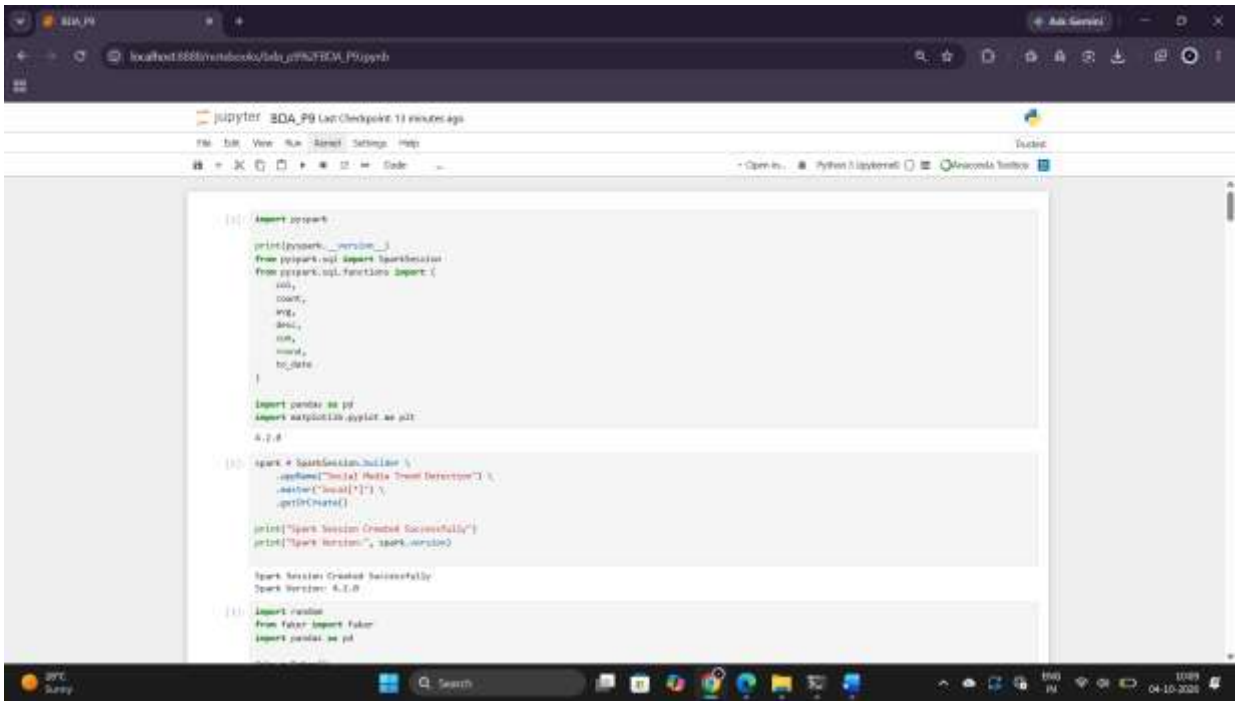
The practical includes creating a SparkSession, loading and displaying the social media dataset, examining the schema, validating the dataset, counting hashtag frequency, identifying top trending hashtags, analyzing hashtag popularity by category and region, comparing trends across multiple days, generating a trend analysis report, and visualizing the results.

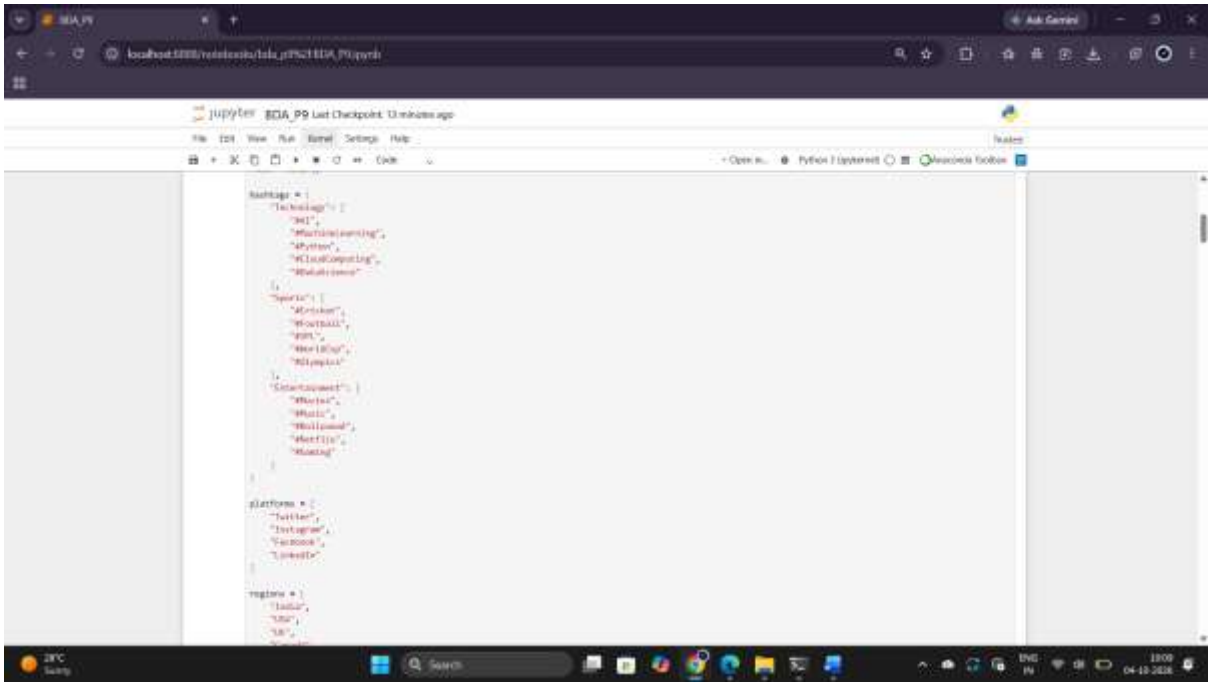
Objectives

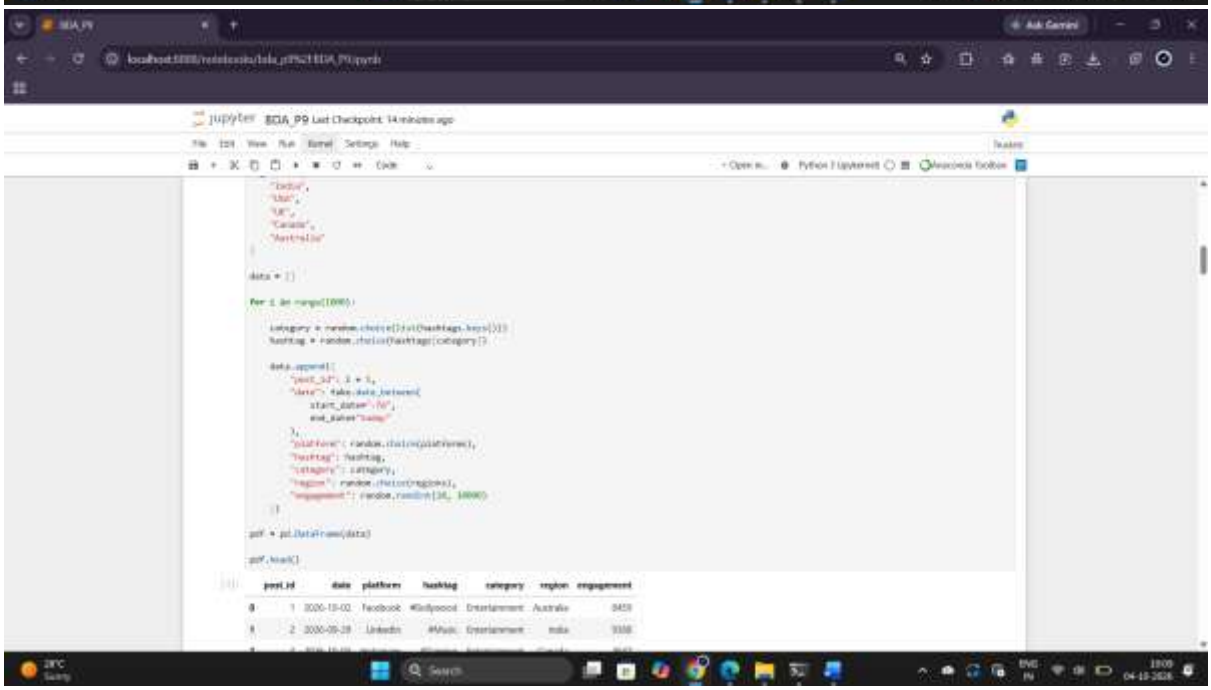
- Understand social media analytics using Apache Spark.
- Create and configure a SparkSession.
- Create or obtain a social media hashtag dataset.
- Load social media records into a Spark DataFrame.
- Display and examine the social media dataset.
- Display the schema of the DataFrame.
- Validate the dataset for missing values and incorrect records.

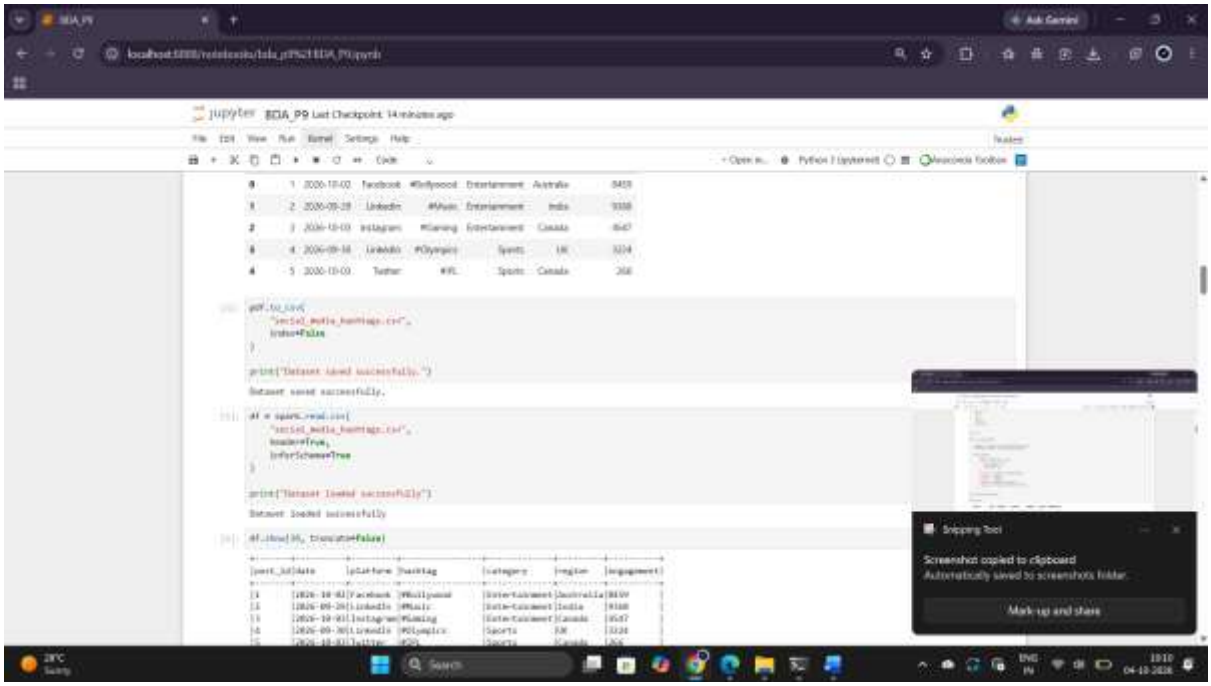
- Calculate hashtag frequency.
- Identify the most popular hashtags.
- Identify the top 3 trending hashtags.
- Analyze hashtag trends by category.
- Compare Technology, Sports, and Entertainment trends.
- Analyze hashtag popularity by region.
- Compare hashtag trends across multiple days.
- Analyze hashtag engagement.
- Calculate a trend score using frequency and engagement.
- Generate a social media trend analysis report.
- Visualize hashtag trends using charts.
- Understand the importance of Spark DataFrame operations in social media analytics.
- Analyze emerging social media trends using PySpark.

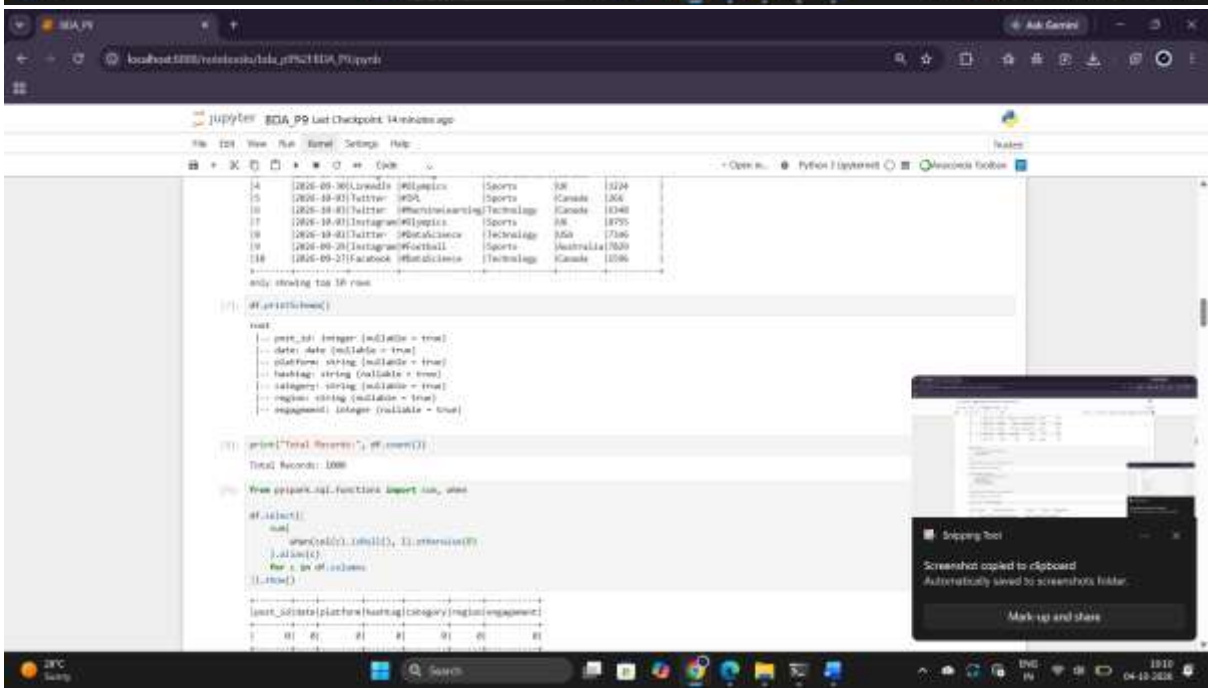
Output

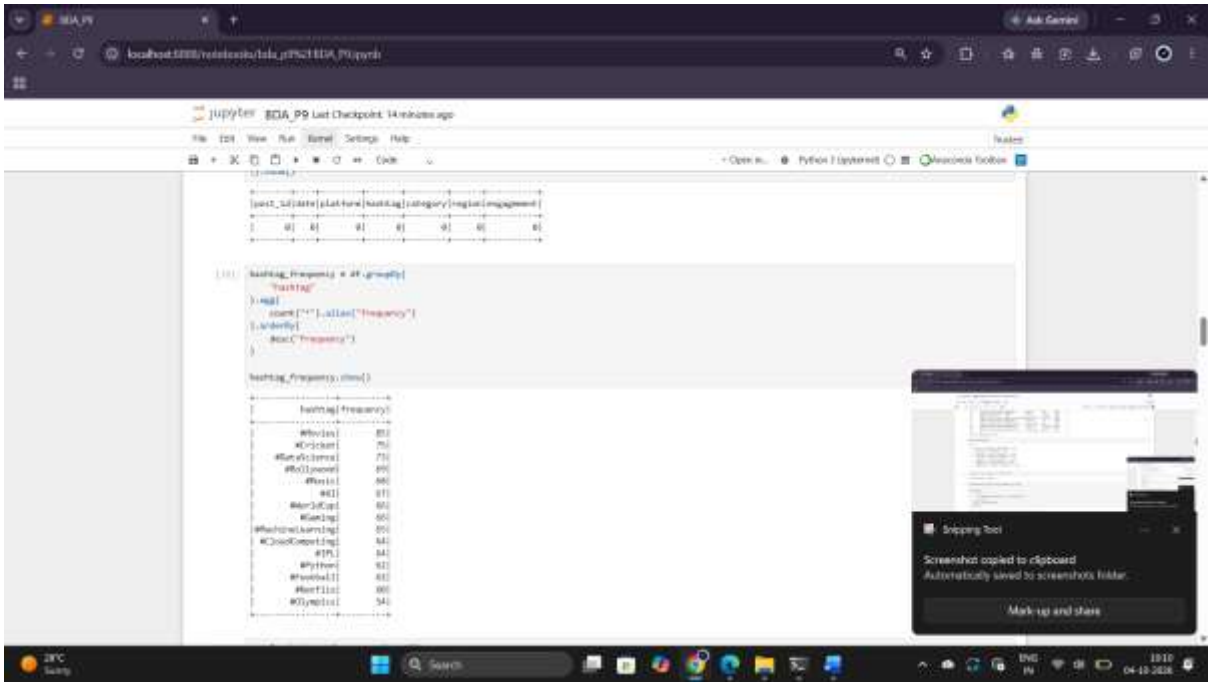


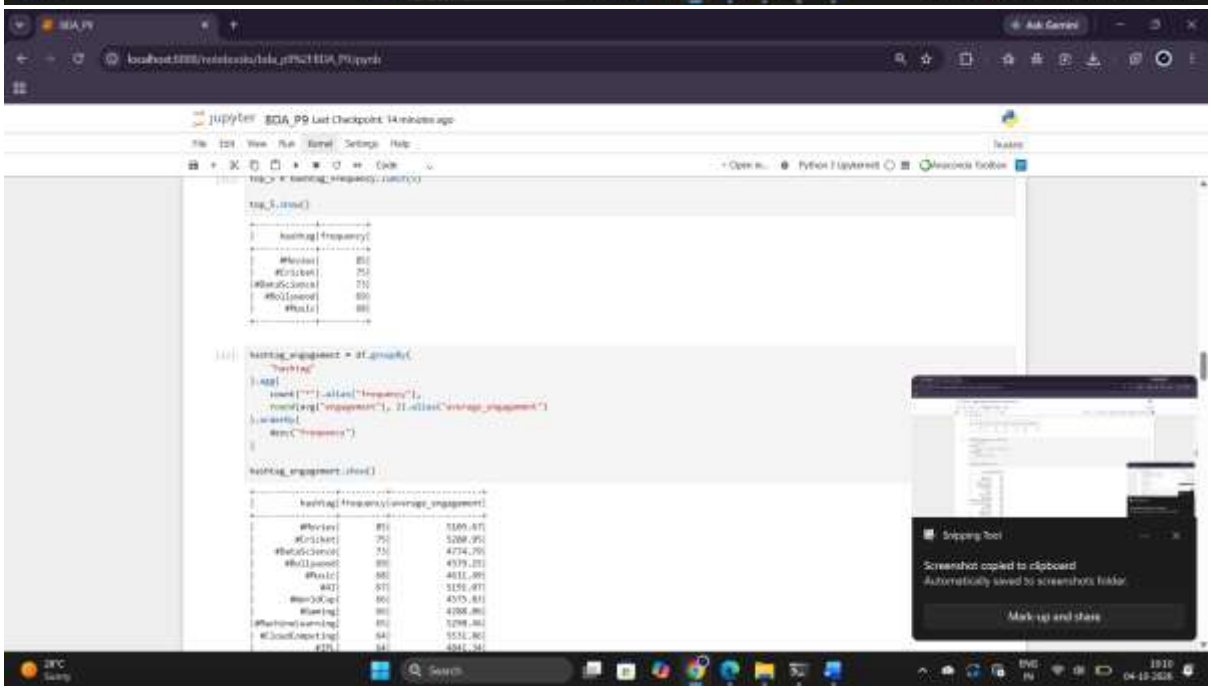


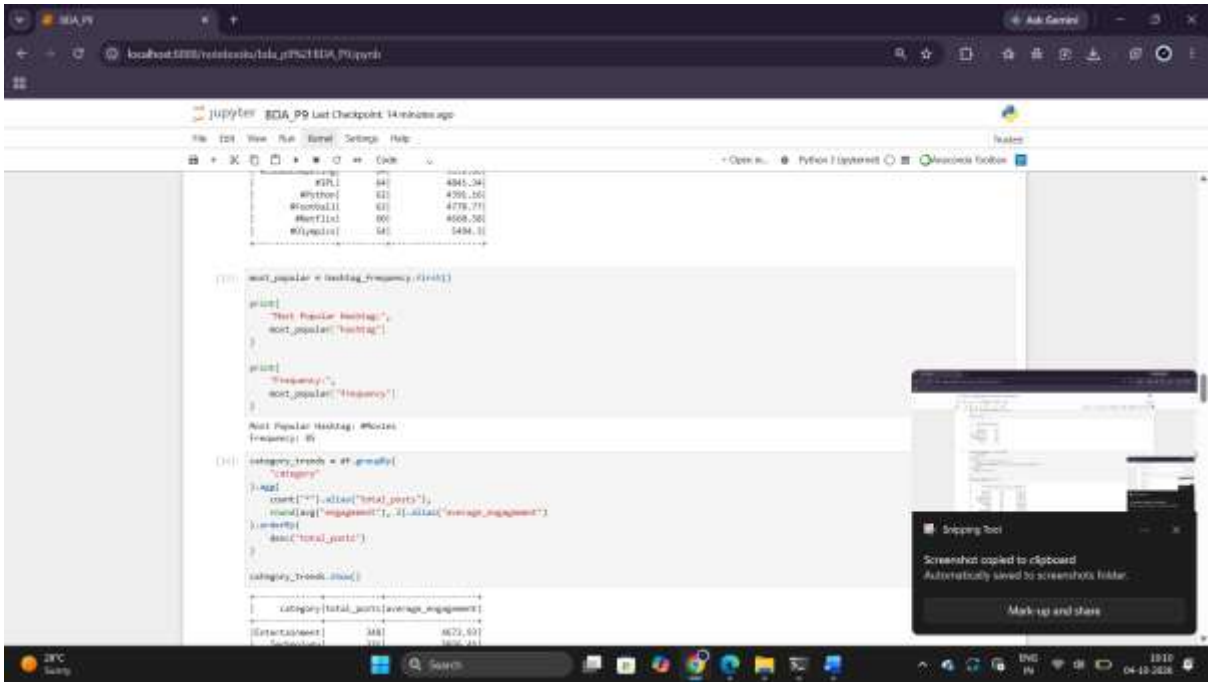


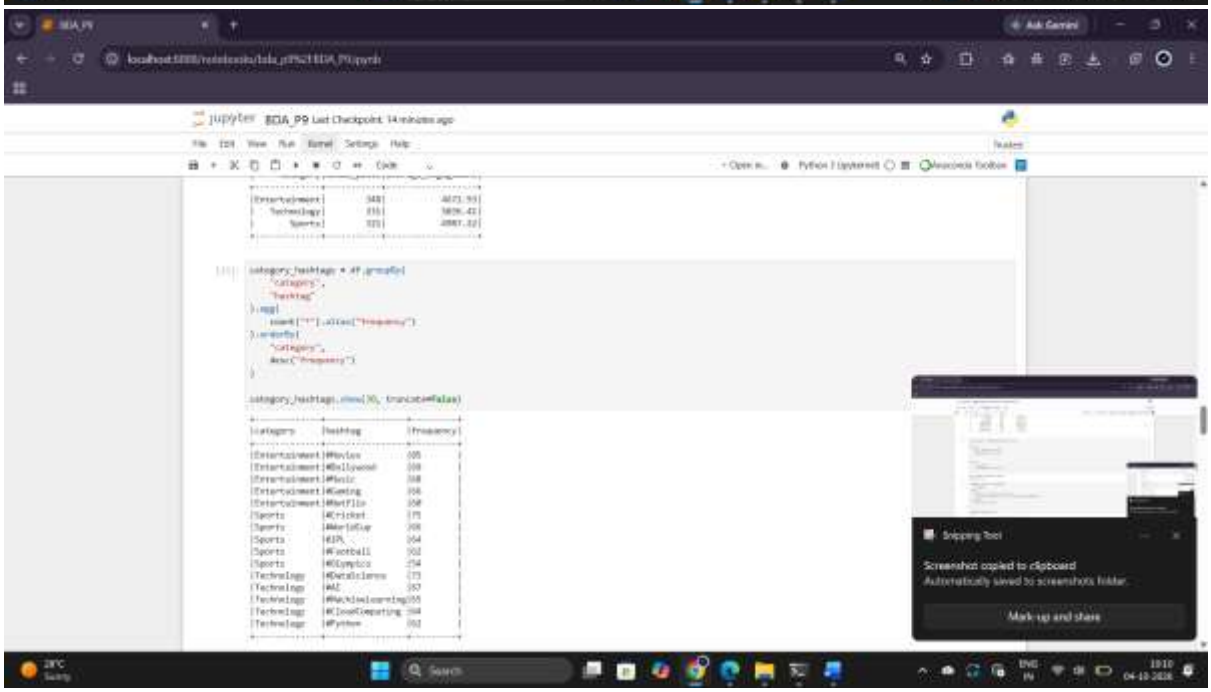


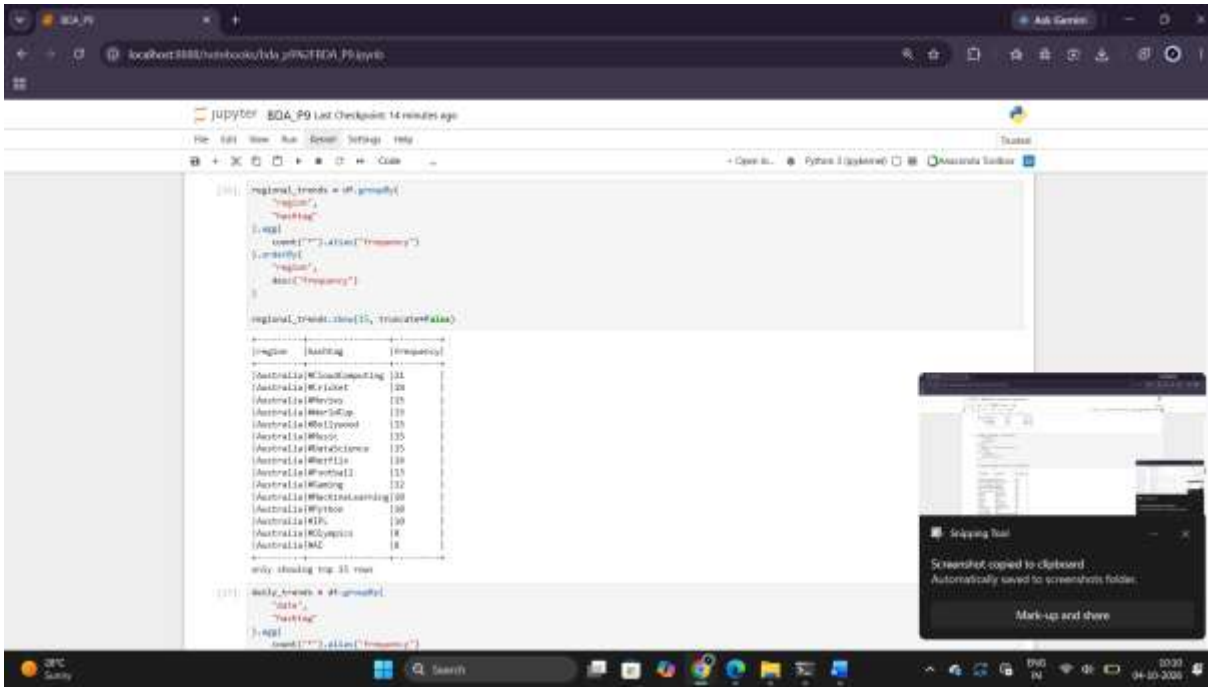


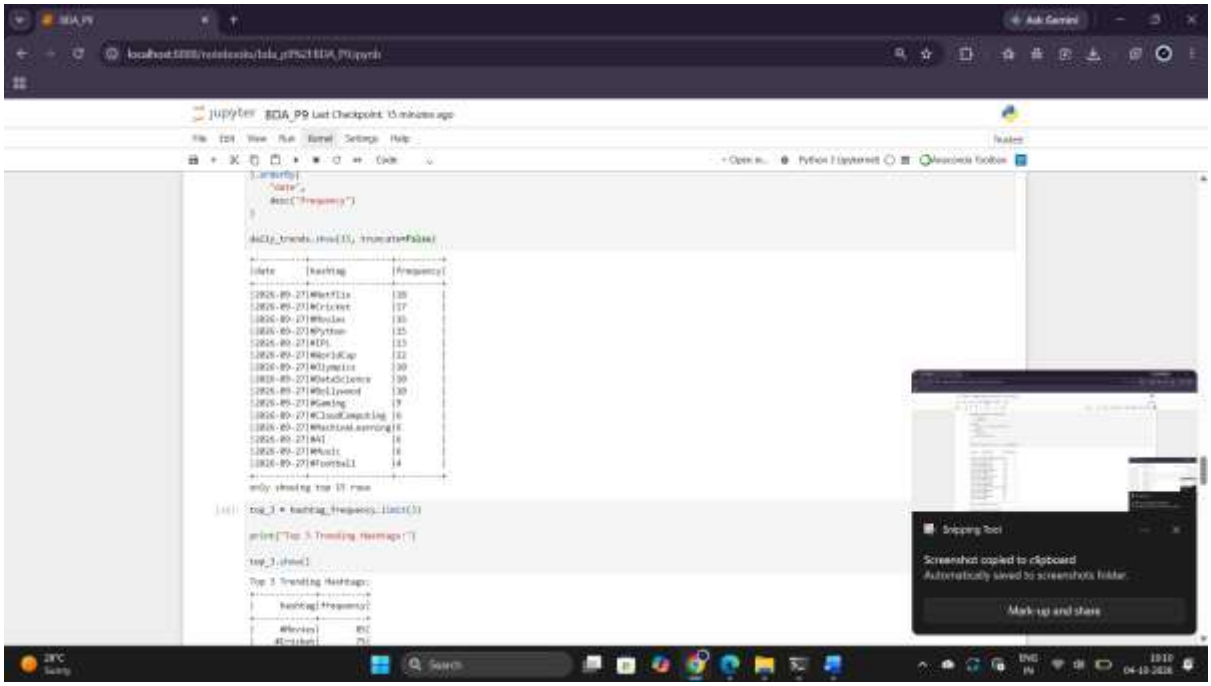

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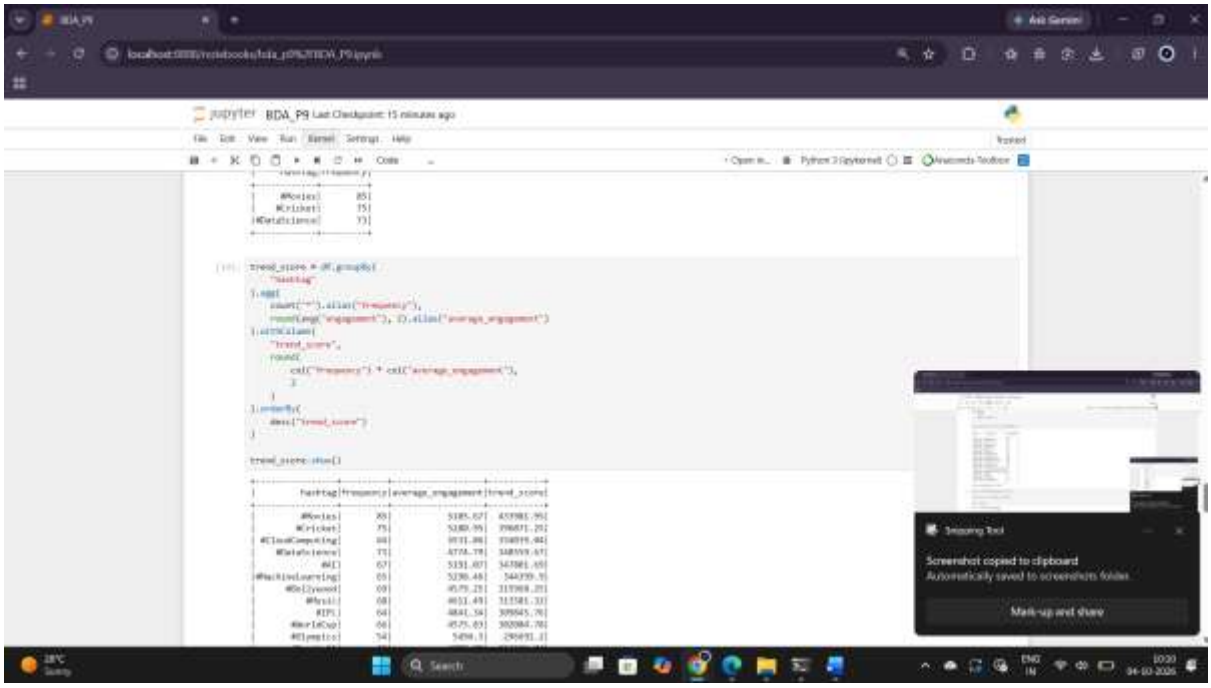

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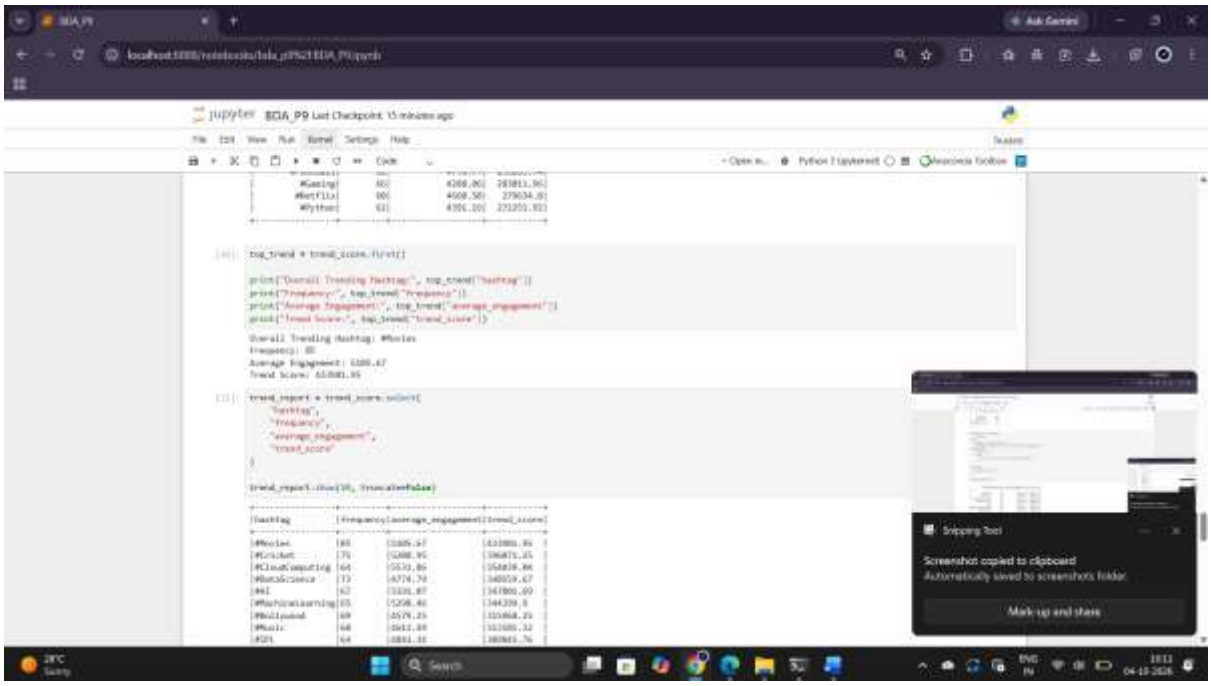


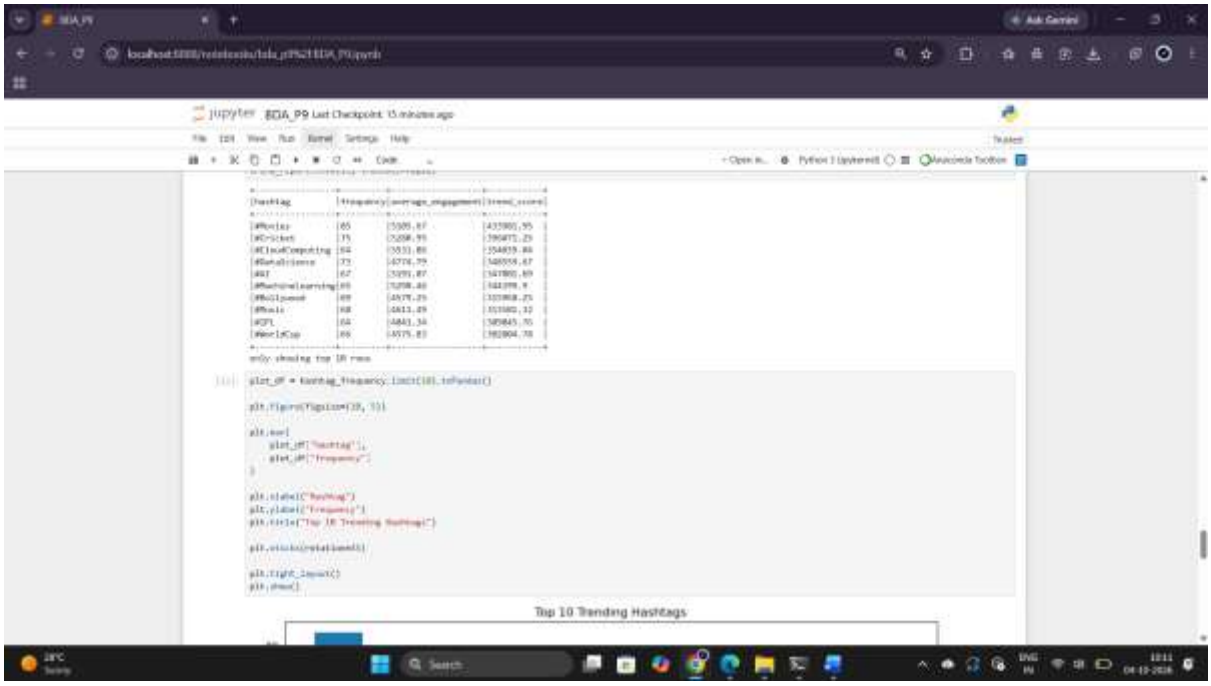


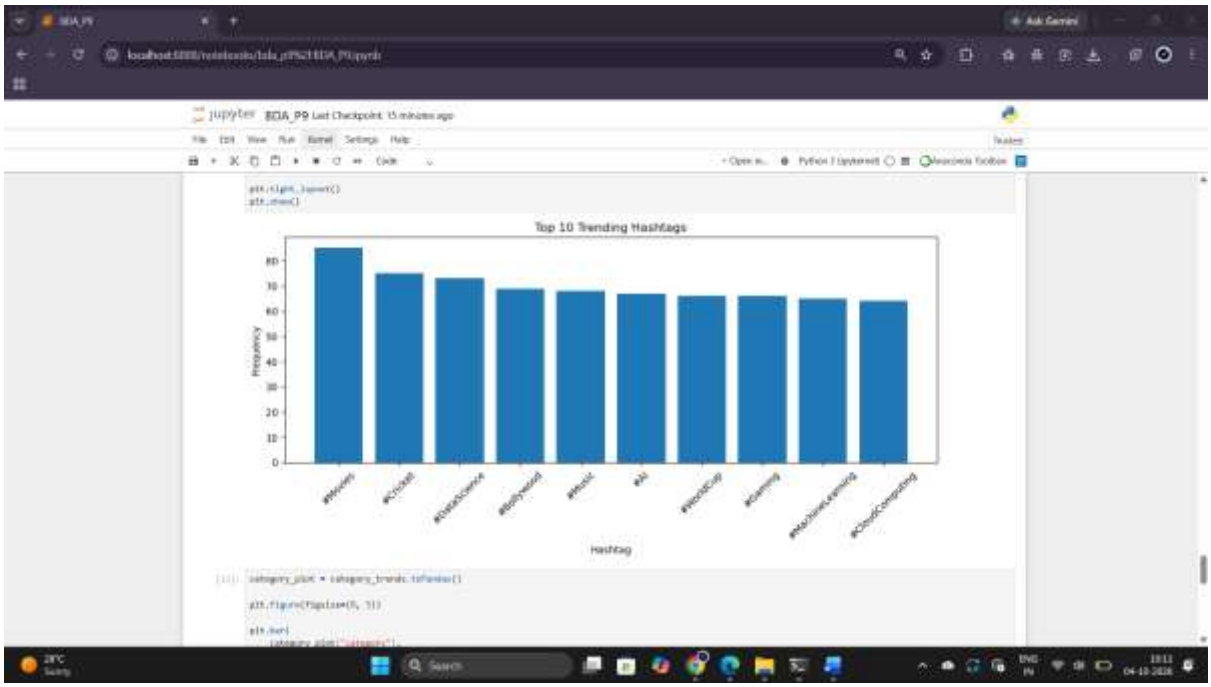


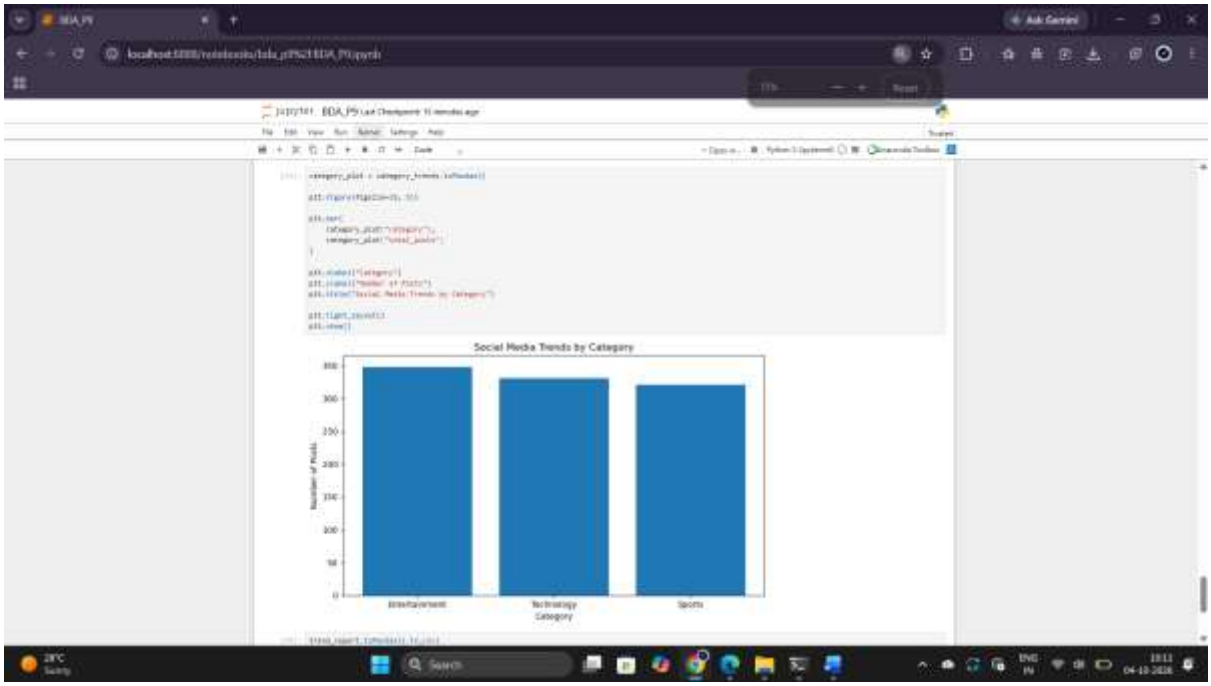


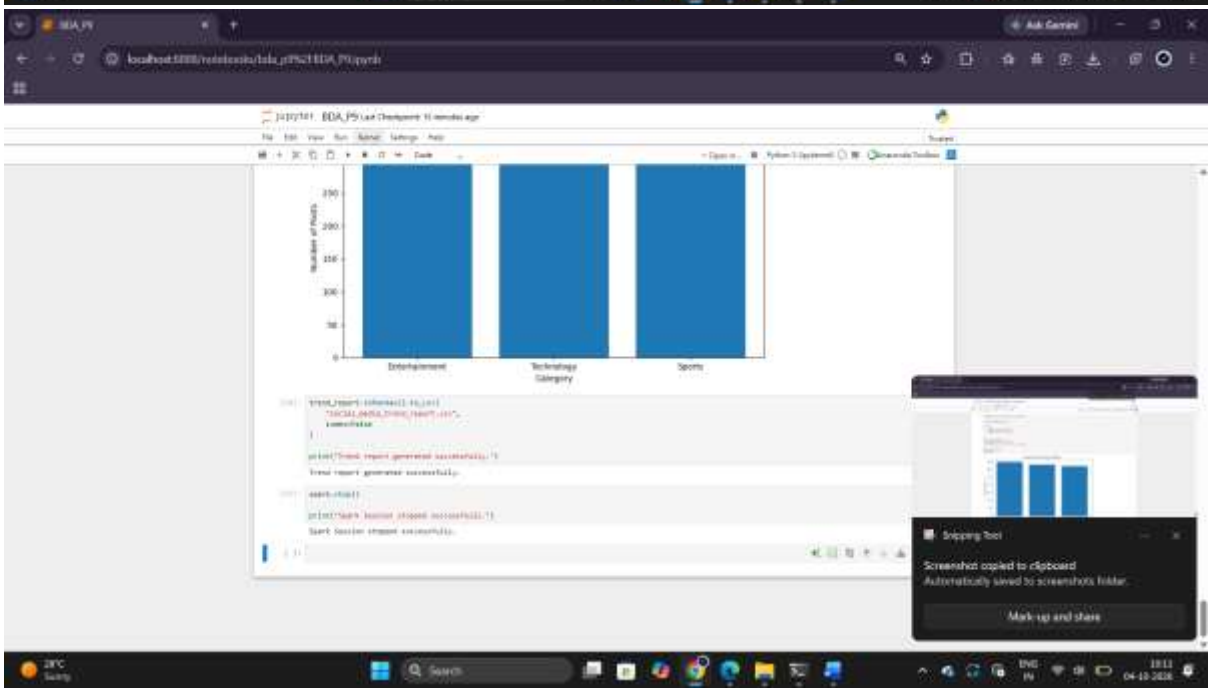

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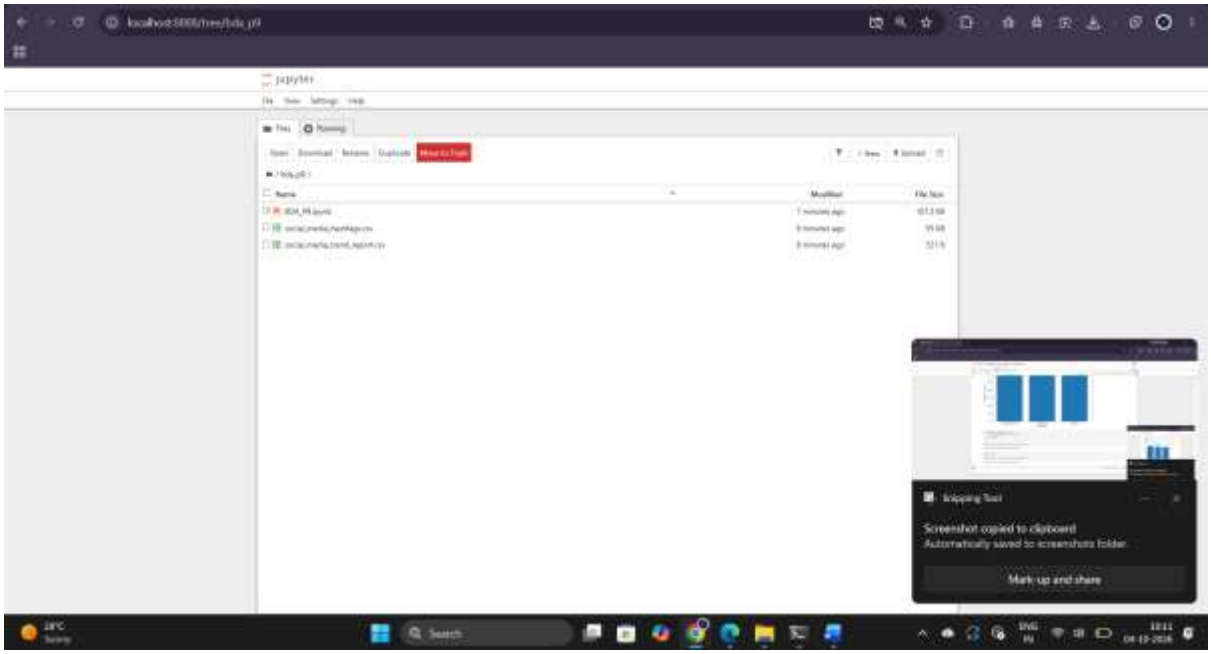

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Top 10 Trending Hashtags


Top 10 Trending Hashtags
Frequency
Hashtag


Social Media Trends by Category
Number of Posts
Entertainment
Technology
Sports
Category


Number of Posts
Entertainment
Technology
Sports
Category
Snipping Tool
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Key Questions / Analysis

Q1. Why is Apache Spark suitable for social media analytics?

Apache Spark is suitable for social media analytics because social media platforms generate very large volumes of data continuously.

Spark can process large datasets efficiently using distributed and parallel processing. Spark DataFrames provide convenient operations for filtering, grouping, aggregation, and analysis of social media records.

Spark can perform operations such as:

1. Hashtag frequency calculation
2. Engagement analysis
3. Category-wise analysis
4. Regional trend analysis

5. Daily trend analysis
6. Identification of popular hashtags
7. Trend detection

Spark can therefore help businesses analyze large-scale social media data efficiently and identify important trends.

Q2. What is the significance of hashtag frequency?

Hashtag frequency represents the number of times a particular hashtag appears in the social media dataset.

A higher hashtag frequency generally indicates that the topic is receiving greater attention or is being discussed by more users.

For example, if:

#AI 150 posts

#Cricket 120 posts

#Movies 95 posts

then #AI has the highest frequency in the dataset.

Hashtag frequency helps businesses:

1. Identify popular topics.
2. Detect trending subjects.
3. Understand user interests.
4. Compare different topics.
5. Plan relevant marketing campaigns.

In PySpark, hashtag frequency can be calculated using:

df.groupBy("hashtag").count().show()

Q3. How can trend analysis support digital marketing campaigns?

Trend analysis helps digital marketers understand what topics and hashtags are currently receiving attention from users.

Businesses can use trending information to create relevant content and marketing campaigns.

Trend analysis can help marketers:

1. Identify popular hashtags.
2. Understand customer interests.
3. Select suitable topics for campaigns.
4. Improve social media engagement.
5. Identify emerging trends.
6. Target specific regions or audiences.
7. Respond quickly to changes in user interests.

For example, if a technology-related hashtag becomes highly popular, a technology company can create social media content related to that topic.

Q4. How do businesses monitor social media trends?

Businesses monitor social media trends by collecting and analyzing data from different social media platforms.

They can analyze:

- Hashtags
- Posts
- Likes
- Comments
- Shares
- Views
- Engagement
- Mentions
- Categories
- Regions
- Trends over time

Apache Spark can process large amounts of social media data and perform aggregation operations to identify popular hashtags and emerging topics.

Businesses can then use these results for marketing, customer analysis, product planning, and decision making.

Q5. How can real-time analytics improve decision making?

Real-time analytics allows businesses to analyze incoming social media data as it is generated.

Instead of waiting for a large batch of data to be processed, businesses can identify important changes quickly.

Real-time analytics can help businesses:

1. Detect emerging trends.
2. Monitor customer reactions.
3. Identify viral content.
4. Track hashtag popularity.
5. Detect sudden changes in engagement.
6. Respond quickly to customer feedback.
7. Improve marketing decisions.
8. Monitor campaigns in real time.

For real-time social media applications, Spark Structured Streaming can be used to process continuously arriving data and perform analytics as new records are received.

Supplementary Problems

1. Identify top 3 trending hashtags.
2. Compare trends across multiple days.
3. Analyze hashtag popularity by region.
4. Create hashtag categories (Technology, Sports, Entertainment).
5. Generate trend visualization using charts.

Conclusion

The Social Media Trend Detection Using PySpark practical was successfully completed using Apache Spark and PySpark.

A social media hashtag dataset was created/loaded into a Spark DataFrame, and its records and schema were examined. The dataset was validated and hashtag frequency was calculated using Spark DataFrame aggregation operations.

The hashtags were analyzed according to their frequency, category, region, and date. The top trending hashtags were identified, and trends across multiple days were compared. Technology, Sports, and Entertainment categories were also analyzed to understand the popularity of different types of social media content.

The practical demonstrated how Spark DataFrames, groupBy(), count(), aggregation functions, filtering, sorting, and distributed processing can be used for large-scale social media analytics and trend detection.

The analysis can help businesses identify popular topics, understand user interests, support digital marketing campaigns, and improve decision making using social media data.